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Coffee Leaf Disease Classification Report

Report

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Contents

List of Figures	3
List of Tables	4
1 Abstract	5
2 Introduction	5
3 Dataset Description	5
4 Preprocessing and Experimental Setup	6
5 Model 1: MobileNetV2 Transfer Learning	6
6 Model 2: Simple CNN	7
7 Results	8
8 Model Comparison and Discussion	9
9 Limitations and Improvements	9
10 Conclusion	10
References	11

List of Figures

List of Tables

1. Abstract

In this report, two image classification models trained on the same coffee leaf disease dataset are compared. The objective of this model is to classify images of coffee leaves into one of three classes – healthy leaves, coffee leaf rust, and Phoma disease. The models tested in this experiment are a MobileNetV2 transfer learning model and a Simple CNN. Both models use the same training, validation, and testing folders and the same batch sizes and input images of 224 x 224 pixels. The three outputs classes for both models are the same as well. The Simple CNN model performed better in terms of test accuracy, achieving 95.88%, compared to 94.43% in the case of the MobileNetV2 model. However, in terms of the training time, the Simple CNN model needed 82 minutes for training while MobileNetV2 needed only 16 minutes for the frozen-base model run.

2. Introduction

Coffee leaf diseases may decrease crop productivity and quality. Machine vision-based computer models can assist farmers with automatic disease detection by learning from labeled images of sick leaves. In the current project, the task was not only to build a model that can perform classification accurately, but also to compare two methods and analyze the differences between the chosen models. Such analysis can be useful since the model with the best accuracy is not necessarily the most efficient one to deploy. In this paper, the criteria for comparison include such parameters as accuracy, loss, performance per class, the number of model parameters, preprocessing, limitations and possible problems with the dataset.

Two models discussed in this experiment belong to two widely-used types of image classification techniques. The first one uses transfer learning based on pre-trained MobileNetV2. Only a classification head will be trained while the remaining feature extraction part of the model is kept frozen. The second approach uses a custom CNN architecture which will have to learn all convolutional filters on the fly during the training process using the dataset. Both of these models are applied depending on the conditions: transfer learning in case there is a lack of labeled data and vice versa.

3. Dataset Description

The external dataset mentioned in the notebook is “A Machine Learning Dataset for Classification of Common Coffee Leaf Diseases in Uganda.” The dataset is posted on Mendeley Data with DOI: 10.17632/k36wnd6knb.1 [1]. According to the source page, the dataset contains a set of labeled images of coffee leaf diseases collected under various lighting conditions on farms in Uganda. These images are classified into three classes: health leaves, leaf rust, and Phoma disease [1]. Images are stored as JPEG files of 256x256 pixels, whereas the notebooks load the images at 224x224 pixels prior to the training [1].

An important point about reproducibility of this dataset should be noted. According to the description,

the Mendeley source provides 3,312 images, but the notebook discovered a slightly different number of local files before cleanup. The notebook then cleaned corrupted files and used 2,252 training images, 483 validation images, and 485 test images for the model training in total of 3,220 images. In this report, the external source is used for documenting the dataset origin, and model results are obtained with the exact numbers provided by the uploaded notebooks.

Table 1. Count of usable images printed by the transfer-learning notebook after image cleansing.

Class	Training images	Validation images	Test images	Total usable images
Health leaves	754	163	162	1,079
Leaf rust	721	154	156	1,031
Phoma	777	166	167	1,110
Total	2,252	483	485	3,220

Figure 1. Count of images per class and split after image cleansing, as generated by the notebook.

4. Preprocessing and Experimental Setup

The images were preprocessed and prepared for training in the same way for both models. The images were read from folders and scaled to 224 x 224 pixels. They were batched with batch size 32 and transformed to categorical labels. The training set was augmented with random rotations, width shifts, height shifts, horizontal flips, and zooming. Validation and test data were not augmented; they were only rescaled.

Models were compiled with categorical cross-entropy as a loss function and Adam optimizer with the initial learning rate of 0.001. The early stopping strategy, reduce learning rate on plateau, and save best model were used during model training. As documented by Keras, EarlyStopping is a callback that stops model training when no improvement in the monitored quantity was observed. Similarly, ReduceLROnPlateau is a callback that decreases learning rate upon a lack of improvement during training [5], [6]. Finally, the classification reports are generated with scikit-learn using its *classification_report* function [7].

One methodology error needs to be fixed in the future run. According to the MobileNetV2 documentation in Keras, rescaling to [0..1] interval is not recommended. The documentation recommends using the MobileNetV2 preprocessing function to rescale inputs to [-1..1] interval. Even though the model achieved very good results, its preprocessing pipeline needs to be updated for a clean transfer-learning experiment.

5. Model 1: MobileNetV2 Transfer Learning

Model 1 applied MobileNetV2 for feature extraction with transfer learning. Specifically, MobileNetV2 was proposed as an efficient mobile network with inverted residuals and linear bottlenecks [2]. Similarly, the Keras application guide mentions that MobileNetV2 applies inverted residual blocks with bottlenecks and supports ImageNet pre-trained weights [3].

The notebook uploads the MobileNetV2 base with ImageNet weights, removes the original top classifier, and freezes the base. The new classification head contains global average pooling, a 0.3 rate dropout, and a dense softmax classifier with three output classes. Thus, the workflow matches the typical transfer learning pipeline as follows: load pre-trained base model, freeze base model, create and stack the new model on top, and then train the classifier on the target data [4].

Table 3. Summary of the MobileNetV2 transfer learning model.

Component	Description
Base model	MobileNetV2; include_top=False; ImageNet weights
Training status of the base model	Frozen during the main training run
Pooling layer	GlobalAveragePooling2D
Dropout	0.3
Classifier output layer	Dense layer with 3 softmax outputs
Total parameters	2,261,827
Trainable parameters	3,843
Non-trainable parameters	2,257,984
Number of epochs with frozen base model	20 epochs

Additionally, the notebook provides an optional fine-tuning stage where the top layers of MobileNetV2 are unfrozen and trained with a reduced learning rate. However, the results comparison table only includes the primary MobileNetV2 model evaluation score of 94.43% test accuracy. Since the notebook does not include a separate final test evaluation result after the fine-tuning block, the following analysis does not treat the fine-tuned model as a separate third evaluation result.

Figure 2. Training and validation accuracy and loss curves for MobileNetV2 with frozen base model.

6. Model 2: Simple CNN

Model 2 is a custom convolutional neural network trained from scratch. In contrast to Model 1, Model 2 does not use pre-trained ImageNet weights. The architecture consists of several convolutional blocks with Conv2D layers, batch normalization, max pooling, and dropout. The last classifier contains dense layers with three class softmax outputs. Therefore, this design enables Model 2 to extract features from the coffee leaf images directly. Still, the training time will be more extended compared to MobileNetV2 since all trainable filters need to be learned from scratch using the project dataset.

Table 4. Summary of the Simple CNN model.

The training curves demonstrate gradual improvement and slightly unstable validation loss in the initial training epochs. It is natural for a CNN model trained from scratch due to randomly initialized network weights. Once the network learns useful visual filters from the input images, validation

Component	Description
Architecture type	Custom CNN trained from scratch
Input image size	224 x 224 x 3
Layers	Conv2D, BatchNormalization, MaxPooling2D, Dropout
Classifier output layer	Dense layer with 3 softmax outputs
Total number of parameters	1,442,851
Training status of all model parameters	All model weights trained from the coffee leaf data
Number of epochs completed	22 epochs
Total training time	4,919.10 seconds / 81.99 minutes

accuracy increases, and the model restores the best performance weights according to the callback settings.

Figure 3. Training and validation accuracy and loss curves for Simple CNN model.

7. Results

The same hold-out test set of 485 images was used for both models. This is an important criterion for comparability since both models' performance was measured against the same image distribution and class labels. The Simple CNN model yielded the highest overall test accuracy and lowest test loss value in this run.

Table 5. General comparison between both models based on the uploaded notebook outputs.

Metric	MobileNetV2 transfer learning	Simple CNN	Best result
Test accuracy	94.43%	95.88%	Simple CNN
Test loss	0.1630	0.1158	Simple CNN
Total parameters	2,261,827	1,442,851	Simple CNN smaller total model
Trainable parameters in main run	3,843	1,442,851	MobileNetV2 fewer trainable parameters
Training time	969 seconds / 16.15 minutes	4,919.10 seconds / 81.99 minutes	MobileNetV2 faster training
Model size in notebook summary	8.63 MB	5.50 MB	Simple CNN smaller file/model summary
Training strategy	Transfer learning	From scratch	Depends on objective

Table 6. Precision, recall, and F1-score per class in the notebooks' classification reports.

Class	MobileNetV2 precision	MobileNetV2 recall	MobileNetV2 F1	Simple CNN precision
Health leaves	0.95	1.00	0.97	0.99
Leaf rust	0.89	0.97	0.93	0.91
Phoma	1.00	0.87	0.93	0.98
Macro average	0.95	0.95	0.94	0.96
Weighted average	0.95	0.94	0.94	0.96

Figure 4. Confusion matrix of MobileNetV2 on the test set. Figure 5. Confusion matrix of Simple CNN on the test set.

8. Model Comparison and Discussion

The Simple CNN yielded the highest numerical score in this run, achieving a test accuracy of 95.88%, compared to 94.43% by MobileNetV2. The gap between the two is 1.45 percentage points. Furthermore, the Simple CNN model also improved the recall value of the Phoma label from 0.87 to 0.92. This is an improvement of 0.05, which indicates that Simple CNN identified less false-negative Phoma instances than MobileNetV2. This is relevant since disease identification should aim to minimize false negatives.

However, one cannot conclude that Simple CNN is automatically the superior model. The Simple CNN required approximately 82 minutes for training, compared to around 16 minutes for MobileNetV2. This is because only 3,843 parameters were trainable in the main run of MobileNetV2, compared to 1.44 million for the Simple CNN model. Training time is an important criterion when quick model testing is crucial, thus making MobileNetV2 more efficient.

The confusion matrix results revealed that both models can effectively classify healthy leaves, as seen by their high values in the True Positive and True Negative cells. However, MobileNetV2 could not identify all healthy test samples but classified 18 images labeled as Phoma into the leaf rust category. In contrast, the Simple CNN made mistakes when identifying healthy samples but minimized Phoma-to-rust confusion by classifying 12 Phoma samples as leaf rust instead. Thus, Simple CNN seemed to better distinguish the two classes in this run.

Finally, one must take care in interpreting the result since the dataset source mentioned that augmentation techniques were utilized to improve sample diversity [1]. Augmented versions of the same leaves might have been included in each split, leading to an overestimated measurement of accuracy. While the notebooks generated random train, validation, and test folders, there was no evidence that the data were split in a manner that ensures each sample group contains similar augmented variants.

9. Limitations and Improvements

The requirement for the assignment is fulfilled since both models are trained and tested on the same dataset with coffee leaves. There are some suggestions that can be done to improve the scientific nature of the report:

Apply the official MobileNetV2 preprocessing pipeline by using replace the current rescale = 1/255 with preprocess_input() function, then train and test the model once again.

Evaluate the fine-tuned MobileNetV2 model separately on the test set. Although the notebook demonstrates how to train the fine-tuning block of the MobileNetV2 model, there is no evaluation of its results.

Run the code using several random seeds and report the average accuracy and average F1-score.

Ensure that there is no data leakage when splitting the datasets into training, validation, and test sets. Augmented images or duplicates should not appear in other sets besides the test one.

Evaluate the models on an additional dataset or use images that were collected during the project.

Estimate the inference time per image and memory consumption on the targeted device. Time for training alone cannot be the determinant for selecting a particular model.

Add an explanation technique such as Grad-CAM or others to ensure that the models learn features from the disease regions and not background artifacts. **10. Conclusion**

This report provides a good comparison of two models on a specific dataset containing coffee leaves diseases. The Simple CNN model gave the best test performance results with 95.88
94.43

MobileNetV2 model is trained much faster as only the classifier is trained on the pre-trained base model. The best way to summarize the project is not that one is always the best but each one has its own advantages. The Simple CNN model gives better results in this notebook run, but MobileNetV2 is more efficient during training.

When preparing the report for submission, it is necessary to emphasize that both models are used for the comparison. On the other hand, some limitations should also be taken into account. Thus, among the technical errors, there are using of wrong MobileNetV2 preprocessing, lack of evaluation of the fine-tuned MobileNetV2 model separately, and possibility of the data leak due to image augmentations.

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